MA1220 - Calculus - II

Assignment - I

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Submission Date	:	14/10/24

Problem 1 (Integration).

(a) What values of a and b maximize the value of $\int_a^b (x - x^2)dx$?.

Sol:-

$$\int_a^b (x-x^2)\,dx = \frac{x^2}{2} - \frac{x^3}{3}\Big]_a^b = \left(\frac{b^2}{2} - \frac{b^3}{3} - \frac{a^2}{2} + \frac{a^3}{3}\right)$$

let us find the critical points by setting the derivate wrt a & b to zero

i) wrt a :

$$\frac{d}{da}\left(\frac{b^2}{2} - \frac{b^3}{3} - \frac{a^2}{2} + \frac{a^3}{3}\right) = 0$$

$$\Rightarrow \quad 0 - 0 - \frac{2a}{2} + a^2 = 0$$

$$\Rightarrow \quad a^2 = a$$

$$\Rightarrow \quad a = 0 \;,\; a = 1$$

ii) wrt b

$$\frac{d}{db}\left(\frac{b^2}{2} - \frac{b^3}{3} - \frac{a^2}{2} + \frac{a^3}{3}\right) = 0$$

$$\Rightarrow \quad b - b^2 = 0$$

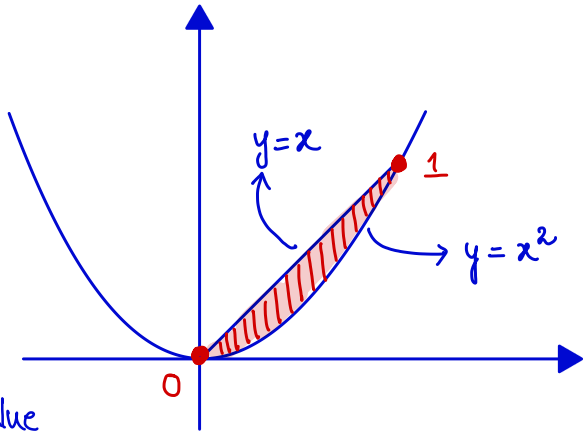
$$\Rightarrow \quad b = 0, 1$$

As a is the lower limit & b is upper limit, $a < b$

Therefore $\boxed{a = 0 \;,\; b = 1}$

$$\Rightarrow \quad I = \int_0^1 (x-x^2)\,dx = \frac{x^2}{2} - \frac{x^3}{3}\Big]_0^1$$

$$= \frac{1}{2} - \frac{1}{3} = \left(\frac{1}{6}\right) \rightarrow \text{max value}$$



Problem 1 (Integration).

(b) What values of a and b minimize the value of $\int_a^b (x^4 - 2x^2)dx$?.

Sol:-

$$\int_a^b (x^4-2x^2)dx = \frac{x^5}{5} - \frac{2.x^3}{3}\Big]_a^b = \left(\frac{b^5}{5} - \frac{2b^3}{3} - \frac{a^5}{5} + \frac{2a^3}{3}\right)$$

let us find the critical points by setting the derivate wrt a & b to zero.

i) wrt a :

$$\frac{d}{da}\left(\frac{b^5}{5} - \frac{2b^3}{3} - \frac{a^5}{5} + \frac{2a^3}{3}\right) = 0$$

$$\Rightarrow \quad 0 - 0 - a^4 + 2a^2 = 0$$

$$\Rightarrow \quad a^2(a^2-2) = 0$$

$$\Rightarrow \quad a = 0 \;,\; a = \sqrt{2} \;,\; a = -\sqrt{2}$$

ii) wrt b

$$\frac{d}{db}\left(\frac{b^5}{5} - \frac{2b^3}{3} - \frac{a^5}{5} + \frac{2a^3}{3}\right) = 0$$

$$\Rightarrow \quad b^4 - 2b^2 - 0 + 0 = 0$$

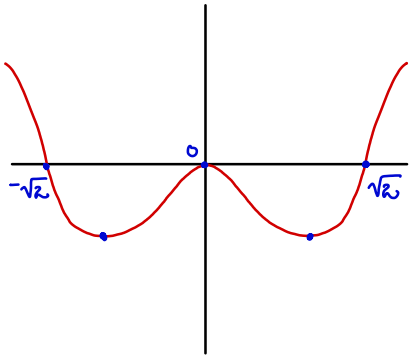
$$\Rightarrow \quad b^2(b^2-2) = 0$$

$$\Rightarrow \quad b = 0 \;,\; b = \sqrt{2} \;,\; b = -\sqrt{2}$$

As a is the lower limit & b is upper limit $\Rightarrow a < b$

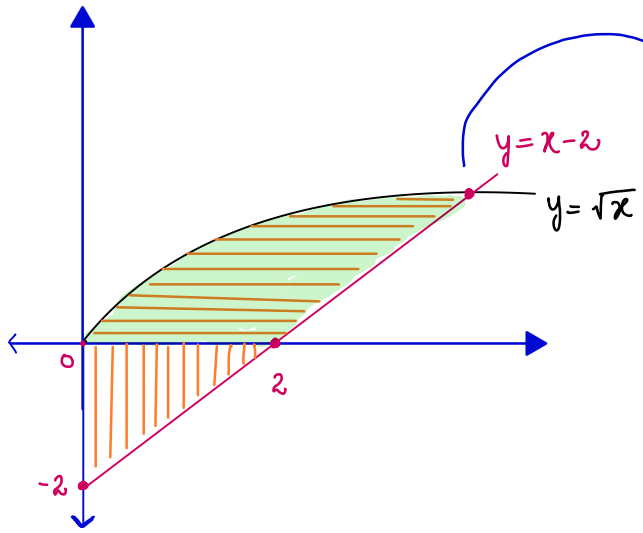
Therefore $a = -\sqrt{2}$, $b = \sqrt{2}$ for min value

$I = \int_{-\sqrt{2}}^{\sqrt{2}} (x^4 - 2x^2)dx$ will give minimum value when $\boxed{a = -\sqrt{2},\; b = \sqrt{2}}$



Problem 2 (Application of integration). Find the area of the region in the first quadrant that is bounded above by $y=\sqrt{x}$ and below by the x-axis and the line $y = x - 2$.

Sol:-



$x-2=\sqrt{x}$

$x^2+4-4x=x$

$x^2-5x+4=0$

$(x-1)(x-4)=0$

$x=1$ or $x=4$

i) If $x=1 \Rightarrow y=x-2=-1$ ✗ (Not in first Quad)

ii) If $x=4 \Rightarrow y=x-2=2$ ✓

$\Rightarrow$ Required Area $= \int_0^4 \sqrt{x}-(x-2)\,dx - Ar(\Delta(0,-2,2))$

$= \left[\frac{2}{3}x^{3/2} - \frac{x^2}{2} + 2x\right]_0^4 - \frac{1}{2}\times 2\times 2$

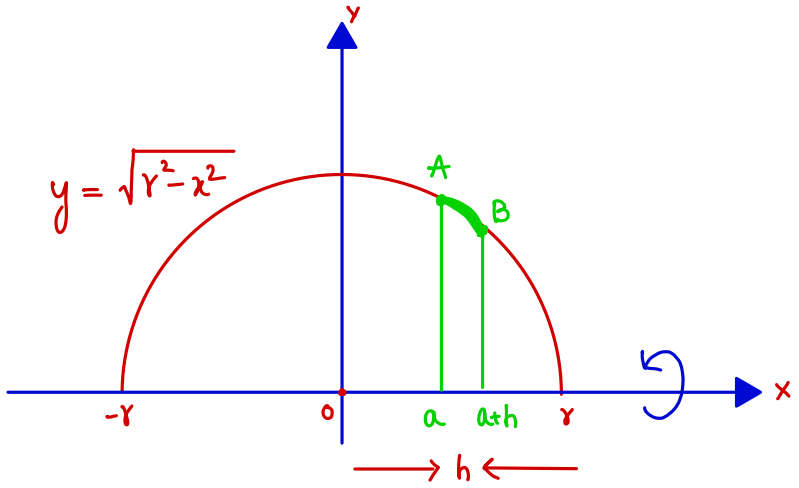
$= \frac{2}{3}(4^{3/2}) - \frac{16}{2} + 2(4) - 2$

$= \frac{16}{3} - 8 + 8 - 2 = \frac{10}{3}$

$\therefore$ Required Area $= \frac{10}{3}$ sq unit

Problem 3 (Surface area). Show that if you cut a spherical loaf of bread into slices of equal width, each slice will have the same amount of crust? Suppose the semicircle $y = \sqrt{r^2 - x^2}$ is revolved about the x-axis to generate a sphere. Let an arc of the semicircle that lies above an interval of length h on the x-axis. Show that the area swept out by the arc does not depend on the location of the interval. (It does depend on the length of the interval.)

Sol :-



The Surface area of a slice is given by $A = 2\pi \int_a^{a+h} y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

for a semicircle $y = \sqrt{r^2 - x^2}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{r^2 - x^2}}(-2x) = \frac{-x}{\sqrt{r^2 - x^2}}$$

$$\Rightarrow A = 2\pi \int_a^{a+h} y \cdot \sqrt{1 + \left(\frac{x^2}{r^2 - x^2}\right)}\, dx = 2\pi \int_a^{a+h} \sqrt{r^2 - x^2} \sqrt{\frac{r^2}{r^2 - x^2}}\, dx$$

$$\Rightarrow A = 2\pi \int_a^{a+h} r \cdot dx = 2\pi r \cdot \left[x\right]_a^{a+h} = 2\pi r\,(a+h - a) = 2\pi r h$$

(Crust)

$\therefore$ We can say that the surface area of each slide will be same as it depends only on width 'h' irrespective of position.

Problem 4 (Improper integral). Discuss the convergence of the following integrals :

(i) $\int_0^\infty \frac{\cos x}{e^x} dx$ = Q

Sol:- i) Let $u = \cos x$, $dv = e^{-x} dx$

$du = -\sin x dx$, $v = -e^{-x}$

$$\int u.dv = uv - \int v.du$$

$$\Rightarrow \int_0^\infty \cos x . e^{-x} dx = \underbrace{-\cos x . e^{-x} \Big]_0^\infty}_{\text{I}} - \underbrace{\int_0^\infty (-e^{-x}).(-\sin x) dx}_{\text{II}}$$

Solving I : $-\cos x . e^{-x} \Big]_0^\infty = \lim_{x\to\infty} (-\cos x . e^{-x}) - (-\cos 0 . e^0)$

$= 0 - (-1.1) = 1$ //

[$\because x \to \infty$, $\cos x$ oscillates b/w 1 to -1 whereas $e^{-x} \to \infty$]

Solving II : $\int_0^\infty e^{-x} . \sin x dx$; $u = \sin x$, $dv = e^{-x} dx$

$du = \cos x dx$ $v = -e^{-x}$

$$\int_0^\infty e^{-x} . \sin x dx = \underbrace{[-\sin x . e^{-x}]_0^\infty}_{\downarrow 0} + \int_0^\infty \underbrace{e^{-x} \cos x dx}_{Q}$$

$= Q$

$$\Rightarrow \int_0^\infty \cos x . e^{-x} dx = \underbrace{-\cos x . e^{-x}\Big]_0^\infty}_{I} - \underbrace{\int_0^\infty (-e^{-x}).(-\sin x) dx}_{II}$$

$$= 1 \qquad = \int_0^\infty \cos x . e^{-x} dx$$

$$\Rightarrow \int_0^\infty \cos x . e^{-x} dx = 1 - \int_0^\infty \cos x . e^{-x} dx$$

$$\Rightarrow \int_0^\infty \cos x . e^{-x} dx = 1/2$$

$$\therefore \int_0^\infty \frac{\cos x}{e^x} dx = 1/2 < 1 \text{ (finite)} \Rightarrow \text{the given improper integral converges.}$$

(ii) $\int_1^\infty e^{x+x^{-1}}\,dx$

Sol:-

Given, $\int_1^\infty e^{x+1/x}\,dx = L_1$

Consider $L_2 = \int_1^\infty e^x\,dx$

we know that, $\int_1^\infty e^x\,dx \to$ diverges $\left[\begin{array}{l} \text{as } x \to \infty \\ e^x \to \infty \end{array}\right]$

and by Comparision,

$$\int e^x dx < \int e^{x+1/x}\,dx$$

$$\Rightarrow \int_1^\infty e^x dx < \int_1^\infty e^{x+1/x} dx \quad [L_2 < L_1]$$

$\downarrow$

as L_2 diverges, we can also say that

$L_1 = \int_1^\infty e^{x+1/x}\,dx$ also diverges

$\therefore$ The given improper Integral diverges.

Problem 5 (Continuity by ϵ and δ)**.** Given the function f and ϵ, show that there exists a δ such that it satisfies the definition of continuity.

(i) $f(x, y) = x^2 + y^2$ with $\epsilon = 0.01$

Sol :-

Any given function $f(x,y)$ is continuos at a point (a,b) if for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $\sqrt{(x-a)^2 + (y-b)^2} < \delta$, then $|f(x,y) - f(a,b)| < \varepsilon$

i) Given $f(x,y) = x^2+y^2$ with $\varepsilon = 0.01$

For $(0,0)$, if $\sqrt{x^2+y^2} < \delta$ then $|f(x,y| < \varepsilon$

$\Rightarrow \quad |x^2+y^2-0| < 0.01$

$\Rightarrow \quad x^2+y^2 < 0.01$

$\Rightarrow \quad (\delta)^2 < 0.01$

$\Rightarrow \quad \boxed{\delta < 0.1}$

$\therefore$ We can say that there exists δ such that it satisfies the definition of continuity at $(0,0)$

Problem 5 (Continuity by ϵ and δ). Given the function f and ϵ, show that there exists a δ such that it satisfies the definition of continuity. at (0,0).

(ii) $f(x,y) = \frac{x+y}{2+\cos x}$ with $\epsilon = 0.02$

Sol :-

ii) Given $f(x,y) = \dfrac{x+y}{2+\cos x}$ with $\varepsilon = 0.02$

for (0,0), if $\sqrt{x^2+y^2} < \delta$ then $\left| \dfrac{x+y}{2+\cos x} - f(0,0) \right| < \varepsilon$

$\Rightarrow f(0,0) = \dfrac{0+0}{3} = 0$

$\Rightarrow \left| \dfrac{x+y}{2+\cos x} - 0 \right| < 0.02$

$\Rightarrow |x+y| < (0.02)(2+\cos x) \longrightarrow (I)$

Using Cauchy- schwarz inequality,

$(a_1^2 + a_2^2 + \cdots a_n^2)(b_1^2 + b_2^2 + \cdots b_n^2) \geq (a_1 b_1 + a_2 b_2 \cdots a_n b_n)^2$

$\Rightarrow (x^2+y^2)(1+1) \geq (x+y)^2$

$\Rightarrow \sqrt{2(x^2+y^2)} \geq |x+y|$

$\therefore$ we can write (I) ; $|x+y| < (0.02)(2+\cos x)$

as $\sqrt{2(x^2+y^2)} < 0.02(2+\cos x)$

Now, $\cos x$ lies between $[-1, 1]$

let us take $\cos x = -1$ for minimum case

$$\Rightarrow \sqrt{2(x^2+y^2)} < 0.02(2-1)$$

$$\Rightarrow 2(x^2+y^2) < (0.02)^2$$

$$\Rightarrow (x^2+y^2) < \frac{4 \times 10^{-4}}{2}$$

$$\Rightarrow \delta^2 < 2 \times 10^{-4}$$

$$\Rightarrow \delta < \sqrt{2} \times 10^{-2}$$

$$\Rightarrow \delta < 1.414 \times 10^{-2}$$

$$\Rightarrow \boxed{\delta < 0.01414}$$

$\therefore$ We can say that there exists δ such that it satisfies the definition of continuity at $(0,0)$

Problem 6 (Continuity). Find the points (x, y) in the plane the function

$$g(x, y) = \frac{x^2 + y^2}{x^2 - 3x + 2}$$

is continuous.

Sol:-

$$\text{Given } g(x,y) = \frac{x^2+y^2}{(x^2-3x+2)} = \frac{x^2+y^2}{(x-1)(x-2)}$$

A function $f(x,y)$ is continuous at a point (x_0, y_0) if

1) f is defined at (x_0, y_0)

2) $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ exists

3) $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0,y_0)$

So, first lets identify where $g(x)$ isn't defined.

$$\Rightarrow g(x,y) = \frac{x^2+y^2}{x^2-3x+2} \longrightarrow x \neq 1,\ x \neq 2$$

$g(x,y)$ isn't defined when $x=1$, $x=2$

ii) ✓ As $g(x) = \frac{\text{Polynomial}}{\text{Polynomial}}$ → always continuous

iii) ✓

$\therefore$ We can say that given $g(x,y)$ is continuous all $(x,y) \in \mathbb{R}^2$ Except $(1,y)$ & $(2,y)$.

Problem 7 (Partial derivative). Find the partial derivative of the function with respect to each variable.

Sol:-

(i) $f(t,\alpha) = \cos(2\pi t - \alpha)$

a) $\frac{\partial}{\partial t} f(t,\alpha) = \frac{\partial}{\partial t} \cos(2\pi t - \alpha)$

$= -\sin(2\pi t - \alpha) \cdot \frac{\partial}{\partial t}(2\pi t - \alpha)$

$= -2\pi \sin(2\pi t - \alpha)$ //

b) $\frac{\partial}{\partial \alpha} f(t,\alpha) = \frac{\partial}{\partial \alpha} \cos(2\pi t - \alpha)$

$= -\sin(2\pi t - \alpha) \cdot \frac{\partial}{\partial \alpha}(2\pi t - \alpha)$

$= \sin(2\pi t - \alpha)$ //

i) $f(t,\alpha) = \cos(2\pi t - \alpha)$

a) $\frac{\partial}{\partial t} f(t,\alpha) = -2\pi \sin(2\pi t - \alpha)$

b) $\frac{\partial}{\partial \alpha} f(t,\alpha) = \sin(2\pi t - \alpha)$

(ii) $g(u,v) = v^2 e^{(\frac{2u}{v})}$

a) $\frac{\partial}{\partial u} g(u,v) = \frac{\partial}{\partial u} v^2 e^{\left(\frac{2u}{v}\right)}$

$= \frac{\partial}{\partial u}(v^2) \cdot e^{2u/v} + \frac{\partial}{\partial u}\left(e^{\frac{2u}{v}}\right) \cdot v^2$ (the $\frac{\partial}{\partial u}(v^2)$ term $\to 0$)

$= e^{2u/v} \cdot \frac{\partial}{\partial u}\left(\frac{2u}{v}\right) \cdot v^2$

$= v^2 \cdot e^{2u/v} \cdot \frac{2}{v} = 2v \cdot e^{2u/v}$ //

b) $\frac{\partial}{\partial v} g(u,v) = \frac{\partial}{\partial v} v^2 e^{\left(\frac{2u}{v}\right)}$

$= \frac{\partial}{\partial v}(v^2) \cdot e^{2u/v} + \frac{\partial}{\partial v} e^{(2u/v)} \cdot v^2$

$= 2v \cdot e^{2u/v} + e^{2u/v} \cdot \frac{\partial}{\partial v}\left(\frac{2u}{v}\right) \cdot v^2$

$= 2v \cdot e^{2u/v} + e^{2u/v} \cdot \frac{-2u}{v^2} \cdot v^2$

$= 2(v-u) \cdot e^{2u/v}$ //

$\therefore$ ii) $g(u,v) = v^2 \cdot e^{2u/v}$

$$\frac{\partial}{\partial u} g(u,v) = 2v \cdot e^{2u/v}$$

$$\frac{\partial}{\partial v} g(u,v) = 2(v-u) \cdot e^{2u/v}$$

(iii) $h(\rho, \phi, \theta) = \rho \sin\phi \cos\theta$

Sol:-

a) $\frac{\partial}{\partial \rho}(h) = \frac{\partial}{\partial \rho} \rho \sin\phi \cos\theta = \sin\phi \cos\theta$ //

b) $\frac{\partial}{\partial \phi}(h) = \frac{\partial}{\partial \phi} \rho \sin\phi \cos\theta = \rho \cos\theta . \cos\phi$ //

c) $\frac{\partial}{\partial \theta}(h) = \frac{\partial}{\partial \theta} \rho \sin\phi \cos\theta = -\rho \sin\phi \sin\theta$ //

Problem 8. Starting from the point $(1,1)$, in which direction one should travel in order to obtain the most rapid rate of decrease of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) = (x+y-2)^2 + (3x-y-6)^2$?

Sol:-

We know that the direction of the most rapid decrease is given by the negative gradient of the function.

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

So,

i) $\frac{\partial f}{\partial x} = 2(x+y-2).1 + 2(3x-y-6).3$

$= 2x+2y-4+18x-6y-36$

$= 20x-4y-40$

ii) $\frac{\partial f}{\partial y} = 2(x+y-2).1 + 2(3x-y-6)(-1)$

$= 2x+2y-4-6x+2y+12$

$= 4y-4x+8$

Now, $\nabla f(x,y) = (20x-4y-40, 4y-4x+8)$

$\Rightarrow \nabla f(1,1) = (20-4-40, 4-4+8)$

$= (-24, 8)$

But for direction of decrease,

$-|f \nabla(1,1)| \Rightarrow -(-24, 8)$

$\equiv (24, -8)$ //

To get the direction independent of the magnitude, we need to give the unit vector in that direction

$$\Rightarrow \sqrt{x^2+y^2} = \sqrt{24^2+8^2} = \sqrt{640} = 8\sqrt{10}$$

$$\Rightarrow \left(\frac{24}{8\sqrt{10}}, \frac{-8}{8\sqrt{10}}\right)$$

$$\therefore \boxed{\text{Unit vector} \equiv \left(\frac{3}{\sqrt{10}}, \frac{-1}{\sqrt{10}}\right)}$$

Problem 9 (Chain rule). Suppose that we substitute polar coordinates $x = r\cos\theta$ and $y = r\sin\theta$ in a differentiable function $w = f(x, y)$. Show that

$$(f_x)^2 + (f_y)^2 = \left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial w}{\partial \theta}\right)^2$$

Sol:- Using chain rule for multivariable functions, we can write that;

i) wrt r : $\frac{dw}{dr} = \frac{\partial w}{\partial x}\cdot\frac{dx}{dr} + \frac{\partial w}{\partial y}\cdot\frac{dy}{dr}$

Now, $x = r\cos\theta$, $y = r\sin\theta$

$\frac{dx}{dr} = \cos\theta$, $\frac{dy}{dr} = \sin\theta$

and $\frac{\partial w}{\partial x} = f_x$, $\frac{\partial w}{\partial y} = f_y$

$$\frac{dw}{dr} = f_x\cos\theta + f_y\sin\theta$$

ii) wrt 'θ' : $\frac{dw}{d\theta} = \frac{\partial w}{\partial x}\cdot\frac{dx}{d\theta} + \frac{\partial w}{\partial y}\cdot\frac{dy}{d\theta}$

$\frac{dx}{d\theta} = r\cdot -\sin\theta$, $\frac{dy}{d\theta} = r\cos\theta$

$$\frac{dw}{d\theta} = r(f_y\cos\theta - f_x\sin\theta)$$

Now, let us solve RHS i.e, $\left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial w}{\partial \theta}\right)^2$

$$\Rightarrow \quad (f_x \cos\theta + f_y \sin\theta)^2 + \frac{1}{r^2}\left[r(f_y \cos\theta - f_x \sin\theta)\right]^2$$

$$= \quad f_x^2 \cos^2\theta + f_y^2 \sin^2\theta + 2 f_x f_y \sin\theta \cos\theta + \frac{1}{r^2} \cdot r^2 \left(f_y^2 \cos^2\theta + f_x^2 \sin^2\theta - 2 f_y f_x \sin\theta . \cos\theta \right)$$

$$= \quad f_x^2 (\cos^2\theta + \sin^2\theta) + f_y^2 (\sin^2\theta + \cos^2\theta) + 2 f_x f_y sc - 2 f_x f_y sc$$

$$= \quad f_x^2 + f_y^2 \qquad \left[\sin^2\theta + \cos^2\theta = 1 \right]$$

$$= \quad \text{LHS}$$

$\therefore$ As LHS = RHS

Hence proved

Problem 10 (Maxima and Minima). Find the triangle for which the product of its three angles is the largest. [You should use the maxima/minima theory of several variables]

Sol:-

Let us consider the angles of a triangle as α, β, γ

we know that $\alpha+\beta+\gamma = \pi$

$$\Rightarrow \quad \gamma = \pi - (\alpha+\beta)$$

Now, the Product $P = \alpha\beta\gamma$

$$= \alpha.\beta\left[\pi - (\alpha+\beta)\right]$$

$$P = \pi.\alpha\beta - \alpha\beta(\alpha+\beta)$$

Let us define $P(x,y) = \pi xy - xy(x+y)$

$$\Rightarrow \quad \nabla P(x,y) = \left(\frac{\partial P}{\partial x}, \frac{\partial P}{\partial y}\right)$$

Now, $\frac{\partial P}{\partial x} = \pi y - \frac{\partial}{\partial x}\left(x^2y + xy^2\right) = \pi y - y.2x - y^2$

$$= y(\pi - 2x - y)$$

and $\frac{\partial P}{\partial y} = \pi x - \frac{\partial}{\partial y}\left(x^2y + xy^2\right) = \pi x - x^2 - x.2y$

$$= x(\pi - 2y - x)$$

To find a critical point,

$$\nabla f(x,y) = 0$$

$$\Rightarrow \frac{\partial P}{\partial x} = 0 \quad , \quad \frac{\partial P}{\partial y} = 0$$

$$\Rightarrow y(\pi - 2x - y) = 0 \quad , \quad x(\pi - 2y - x) = 0$$

$$\Rightarrow \quad y = 0 \ \text{(or)} \quad \pi = 2x + y \qquad x = 0 \ \text{(or)} \quad \pi = 2y + x$$

Here x, y cannot be zeros because triangle cannot be formed

Therefore we must consider

$$\pi = 2x + y, \quad \pi = 2y + x$$

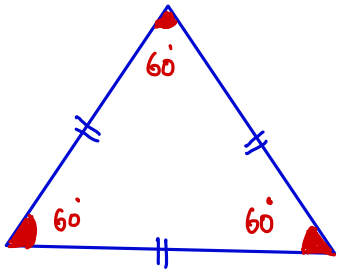
$$\Rightarrow 2x + y = 2y + x$$

$$\Rightarrow x = y$$

$$\& \quad 2\pi = 3(x+y) = 3(2x)$$

$$\Rightarrow x = \pi/3 = y$$

$\therefore$ The 3 angles for which P is maximum are $(\pi/3, \pi/3, \pi/3)$



Equilateral triangle